

Research paper

Hybrid strategy of flexibility regulation and economic energy management in the power system including renewable energy hubs based on coordination of transmission and distribution system operators

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ABSTRACT

Energy management of energy hubs in the power system impacts both transmission and distribution networks and their operations. Hence, to enhance efficiency, there should be bilateral coordination between distribution and transmission networks. Moreover, thanks to including power sources, storage, and responsive loads, hubs can earn financial benefits from the energy and ancillary services market. Therefore, this paper presents the operation of distribution and transmission networks in the presence of flexible-renewable energy hubs based on transmission system operator (TSO)-distribution system operator (DSO) coordination according to the market clearing price modeling in energy and flexibility markets. The proposed scheme minimizes the sum of the expected operating costs of the networks and hubs minus the expected flexibility profit of the hubs. The scheme is constrained to optimal power flow equations of the networks, flexibility limit of the networks, and flexibility-operation model of renewable energy hubs located in transmission and distribution networks. The unscented transformation method is used to model the uncertainties of load, renewable sources, and stationary storage (aggregation of electric vehicles). Energy price and flexibility are among the dual variables of the problem. To simultaneously calculate these variables and the main variables of the problem, an integrated model of the suggested scheme based on the penalty function technique is extracted. Eventually, numerical results validate the capability of the scheme in enhancing the operation, economic, and flexibility status of the proposed scheme, in which renewable sources along with storage and responsive load in the energy hub can obtain 100 % flexibility conditions for the networks. They help enhance the economic and operation indices of distribution and transmission networks by about 26 %-44 % and 22 %-33 % compared to the case without hubs and renewable sources.

1. Introduction

1.1. Motivation

To reduce environmental pollution, the use of renewable energy sources (RESs) in power systems has increased worldwide (Tong et al., 2023). Since the power generation by these sources is uncertain, to have a sustainable energy source, there is a need to install flexibility measures acting as energy storage options and demand response programs (DRPs) to deal with power fluctuations of RESs (Kamwa et al., 2023). In this situation, to improve the functionality of these flexibility sources, their integrated model such as a virtual power plant (VPP) or energy hub (EH) has been proposed in various studies (Bagherzadeh et al., 2024). For

example, an integrated system with RESs and energy storage can manage its flexibility such that its presence in the network does not lead to disruption of network flexibility (Jamali et al., 2019). Also, according to (Moghaddam et al., 2020), an EH as a form of coordination between power sources, energy storage, and responsive loads is able to have a better economic outcome as well as efficiency than the individual management of each element. Additionally, the installation and use of EHs not only is an issue related to the distribution system and the distribution system operator (DSO) but also will affect the transmission network and its operation indeed. The transmission system operator (TSO) and DSO can both benefit from the flexible sources available in the power system. The TSO can use EHs for system balancing, voltage control, congestion management, and other purposes; while the DSO adopts EH to manage its local network issues (AkbariZadeh et al., 2021).

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Nomenclature			
Variables			
Cost	The sum of energy cost minus flexibility profit (\$)	K_{DL}	Incidence matrix of buses and lines in the distribution network
F_{UP}/F_{DW}	Upward/downward flexibility power (MW)	K_{EH}	Incidence matrix of buses and EHs
H_{CH}/H_{DC}	Thermal power storage of thermal energy storage (TES) in charging/discharging modes (MW)	K_{TD}	Incidence matrix of the distribution network and buses of the transmission network
H_{DR}	Thermal power in demand response program (DRP) (MW)	K_{TL}	Incidence matrix of buses and lines in the transmission network
P_{CH}, P_{DC}	Active power of electrical energy storage (EES) in charging and discharging modes (MW)	$\bar{P}_{CH}, \bar{P}_{DC}$	Rate of charge and discharge of the EES (MW)
P_{DL}, Q_{DL}	Active power (MW) and reactive power (MVar) flow through the distribution line	P_D, Q_D, H_D	Active load (MW), reactive load (MVar), and thermal load (MW)
P_{DS}, Q_{DS}	Active power (MW) and reactive power (MVar) flow through the substation	$\underline{P}_{GU}, \bar{P}_{GU}$	Minimum and maximum active power generation by generation unit (MW)
P_{DR}	Active power in the DRP (MW)	P_R	Power generation by RESs (MW)
P_{EH}	Active power of energy hub (EH) (MW)	$\underline{Q}_{GU}, \bar{Q}_{GU}$	Minimum and maximum active power generation by generation unit (MVar)
P_{GU}, Q_{GU}	Active power (MW) and reactive power (MVar) of the generation unit	$\bar{S}_{DS}, \bar{S}_{DL}$	The maximum apparent power of the distribution substation and distribution line (MVar)
P_{TL}, Q_{TL}	Active power (MW) and reactive power (MVar) flow through the transmission line	$\bar{S}_{TL}$	The maximum apparent power of the transmission line (MVar)
V, δ	Magnitude (per unit, p.u.) and angle (rad) of voltage	$\underline{V}, \bar{V}$	Minimum and maximum voltage magnitude (p.u.)
$\bar{\gamma}_{DN}, \underline{\gamma}_{DN}$	Flexibility incentive price (FIP) of the distribution network in upward and downward mode (\$/MWh)	β_F	Flexibility tolerance (MW)
$\bar{\gamma}_{TN}, \underline{\gamma}_{TN}$	FIP of the transmission network in upward and downward mode (\$/MWh)	$\underline{\delta}, \bar{\delta}$	Minimum and maximum voltage angle (rad)
$\lambda_{DN}, \lambda_{TN}$	Energy price (\$/MWh) in distribution and transmission networks	η_{CH}, η_{DC}	The efficiency of charge and discharge of the storage
Constants		η_b, η_l, η_h	The efficiency of the turbine, loss, and thermal power in bio-waste energy unit (BEU)
a_{GU}, b_{GU}, c_{GU}	Coefficients of fuel cost function (respectively in \$, \$/MWh, and \$/MWh ²)	σ	Rate of participation of consumers in the DRP
$\dot{E}_{ES}, \underline{E}_{ES}, \bar{E}_{ES}$	Initial, minimum, and maximum energy of the EES (MWh)	ϑ	The ratio of the thermal power and active power in the BEU
$\dot{E}_{TS}, \underline{E}_{TS}, \bar{E}_{TS}$	Initial, minimum, and maximum energy of the TES (MWh)	Indices	
G_{TL}, B_{TL}	Conductance and susceptance of the transmission line (p.u.)	b	Bus in the distribution network
G_{DL}, B_{DL}	Conductance and susceptance of the distribution line (p.u.)	d	Distribution network
$\bar{H}_{CH}, \bar{H}_{DC}$	Rate of charge and discharge of the TES (MWh)	e	EH
		h	Operation hour
		j	Bus
		n	Bus in the transmission network
		m	Dimension of the uncertain variables vector
		W	Scenario

However, on the one hand, it is not possible to utilize the maximum benefits of EHs in a system where TSOs and DSOs operate their respective networks independently without any coordination, on the other hand in some cases these inconsistencies may cause challenges for both TSO and DSO. Therefore, a coordinated framework for the operation and planning of transmission and distribution systems in the case of EH penetration is essential and very useful for both TSO and DSO. By increasing the level of this coordination, both system operators will be able to effectively and economically support each other in solving the problems of each transmission or distribution network and as a result the whole system (Givisiez et al., 2020). Also, an EH consists of power sources, energy storage, and responsive loads that can benefit these resources by participating in the energy and ancillary services markets. This benefit depends on the optimal energy management of EHs simultaneously in transmission and distribution networks through TSO-DSO coordination.

1.2. Literature review

Various studies have already been conducted on TSO-DSO coordination. Ref (Gupta et al., 2022). presents a new coordinated market design, in which distributed generations (DGs) get involved in both wholesale and local energy markets. A two-stage optimization model is

introduced in (Chen et al., 2021) for an electrical-thermal energy market. In another study, a two-stage market is suggested by (Chen et al., 2022) to provide TSO-DSO coordination in the reactive power ancillary services market. The authors in (Habibi et al., 2022a) present a novel stochastic framework to be applied to the day-ahead and reserve markets while minimizing the level of complexity. Coordination of TSO, DSO, and distributed generation aggregators in a local market has been discussed in (Sheikhahmadi et al., 2021), where a two-stage optimization method is used to solve two subproblems related to distribution operation and market operator. Reference (Habibi et al., 2022b) focuses on presenting a model of scheduling for electrical and thermal energy by taking into account TSO-DSO cooperation such that the efficiency of the network's operation is enhanced. In another research (Jiang et al., 2022), the transactions between energy and flexibility are simplified using a market-clearing model. This is realized by providing coordination between flexibility sources.

Ref (Wang et al., 2022). finds the optimal operation of electric vehicles (EVs) flexibility while considering the energy consumption level of EV owners. At the distribution level, the DSOs analyze the operating range of EV aggregators so that services are properly applied. At the transmission level, the TSO properly schedules the charging plan of EVs and ancillary services. Ref (Steriotis et al., 2022). provides a model of TSO-DSO coordination and presents a new energy service provider so

that renewable energy and storage planning are both optimized. A method of coordination between TSO and DSO is introduced in (Usman et al., 2023) for the day-ahead operating planning of ancillary services, including congestion management and voltage control. A decentralized framework is used in (Bagheri and Jadid, 2022) to clear retail and wholesale markets at the same time by performing technical and economic investigations so that the benefits and shortcomings of the framework compared to a centralized method become evident. Ref (Pearson et al., 2022). discusses various approaches to integrate flexibility sources, i.e., distributed energy resources. The paper also investigates the potential of such resources for enhancing load flexibility. Ref (Mansouri et al., 2023a). introduces a method to determine the necessary adaptability of the TSO through the utilization of distributed energy resources (DERs), dynamic end-users like smart buildings (SBs), and EV fleets at the distribution level. This involves introducing an innovative approach to develop a demand response program (DRP), wherein tariffs are structured with varying incentives over time to align with flexibility requirements. Ref (Gu et al., 2023). outlines a strategy for investing in DERs and energy storage within a coordinated framework involving both TSOs and DSOs. This approach tackles two key facets of DSO flexibility aggregation: enhancing flexibility and providing dynamic flexibility simultaneously. A summary of the above-mentioned surveys is given in Table 1.

1.3. Research gaps

Based on the research background and Table 1, the following major research gaps exist in the field of simultaneous operation of transmission and distribution networks based on TSO-DSO coordination:

- The presence of controllable power sources in a network helps enhance the economic and technical indices of that network compared to power flow studies. Meanwhile, EHs are capable of extracting high energy efficiency due to the simultaneous management of several forms of energy (e.g., electricity and heat). Therefore, it is predicted that their presence in electrical networks and their optimal energy management will lead to the improvement of both distribution and transmission network operations. Nonetheless, the presence of EH has been mentioned in a few research pieces. In (Jiang et al., 2022) and (Bagheri and Jadid, 2022), only the energy management of several distribution networks with different energy

consumptions has been discussed in connection with TSO, but the presence of EHs is not considered.

- Various RESs can play a key role in supplying clean energy (with very low pollution). Among them, the use of wind turbines (WTs) and photovoltaics (PVs) in the power system has gained more interest in the past decade. However, less research has been done in the area of bio-waste energy units (BEU). BEUs play a useful role in reducing environmental pollution by converting waste into energy (Voelklein et al., 2019). Also, BEU first turns waste into gas. Therefore, if the BEU is equipped with a combined heat and power (CHP) system, it can simultaneously produce sustainable electricity and heat. Therefore, it is predicted that its presence in an EH can increase the energy efficiency and flexibility of the EH, and can also realize an EH with low emissions. However, this issue has been mentioned in a few researches works.
- The presence of RESs decreases the network flexibility because of the uncertainty in their power generation. This issue will result in unequal real-time and day-ahead operations, which may lead to an imbalance of generation and consumption in real-time operations (Mansouri et al., 2023b). To compensate for this issue, various studies such as (Gupta et al., 2022; Chen et al., 2021, 2022; Habibi et al., 2022a, 2022b; Sheikahmadi et al., 2021; Jiang et al., 2022; Wang et al., 2022; Steriotis et al., 2022; Usman et al., 2023; Bagheri and Jadid, 2022; Pearson et al., 2022) have mentioned the utilization of flexible sources (elements that can control their active power). Nevertheless, in a few studies such as (Jiang et al., 2022; Wang et al., 2022), flexibility modeling has been proposed.
- Due to the increasing penetration of DERs at the distribution level, TSO-DSO coordination is becoming a crucial issue in power system operation. According to Table 1, in most research, scenario-based stochastic optimization (SBSO) has been used to model the uncertainties. This method requires a significant number of scenarios to obtain a reliable solution. Therefore, the problem dimension and its calculation time will be quite high, especially in large-scale systems. To tackle this issue, some deterministic techniques such as robust optimization have been proposed (Bagheri and Jadid, 2022). However, to evaluate the flexibility, several RES scenarios are needed. Therefore, robust optimization is not suitable for flexibility modeling. To compensate for this challenge, stochastic optimization based on methods with a minimum number of scenarios is required. Yet, this has been discussed in a few studies related to TSO-DSO coordination.

Table 1
Taxonomy of recent research works.

Ref.	Capability of energy hub on TSO-DSO coordination	Flexibility regulation in transmission and distribution network	Economic model of flexibility	CHP-based BEU model	Uncertainty modelling
(Gupta et al., 2022)	x	x	x	x	-
(Chen et al., 2021)	Considering heating network, no energy hub	x	x	x	-
(Chen et al., 2022)	x	x	x	x	-
(Habibi et al., 2022a)	x	x	x	x	Scenario-based stochastic optimization (SBSO)
(Sheikahmadi et al., 2021)	x	x	x	x	-
(Habibi et al., 2022b)	Considering heating network, no energy hub	x	x	x	-
(Jiang et al., 2022)	x	✓	✓	x	SBSO
(Wang et al., 2022)	x	✓	x	x	SBSO
(Steriotis et al., 2022)	x	x	x	x	-
(Usman et al., 2023)	x	x	x	x	SBSO
(Bagheri and Jadid, 2022)	x	x	x	x	Robust optimization
(Pearson et al., 2022)	x	x	x	x	-
(Mansouri et al., 2023a)	x	✓	x	x	SBSO
(Gu et al., 2023)	x	✓	x	x	SBSO
This work	✓	✓	✓	✓	UT-based stochastic optimization

1.4. Contributions

To compensate for the first and third research gaps, in this paper, the simultaneous operation of transmission and distribution networks in the presence of flexible-renewable EHs based on TSO-DSO coordination is presented by considering the energy and flexibility market clearing model, as shown in Fig. 1. The EHs in this scheme are fully based on RESs, and the thermal energy supply source in the EH uses CHP-based BEU, to fill the second research gap. Based on Fig. 1, the wholesale market covers the transmission network, which is managed by the TSO. Moreover, DSOs and some EHs can participate in this market and trade energy. There is also a retail market in the distribution network, which is managed by DSO. Besides, in the wholesale and retail markets, EHs provide flexible ancillary services to adjust the flexibility of the transmission and distribution networks.

The proposed framework minimizes the difference in the total expected energy costs of the TSO, DSO, and EHs, and the sum of the expected flexibility revenue of EHs. This scheme is constrained to the optimal AC power flow and the flexibility constraints in both transmission and distribution networks. Besides, in the proposed model the formulation of the operation-flexibility of renewable EHs is considered based on stationary and mobile storage devices and the DRP. The flexibility constraints in both networks are used to include the flexibility regulation of these networks.

In the proposed model there are several uncertain resources such as the load, RESs, and parameters of mobile energy storage devices (e.g., EVs). To deal with the last research gap, this paper uses a stochastic optimization approach based on the unscented transformation (UT) method to handle the aforementioned uncertainties. In this method, the number of scenarios generated is almost twice the number of uncertain parameters, which has the lowest number of scenarios compared to other stochastic optimization techniques (Kazemi et al., 2022; Dabaghjamesh et al., 2018). Energy and flexibility prices are two

variables of the proposed problem. Therefore, for the simultaneous calculation of these variables and the main variables of the problem, an integrated model for the proposed scheme is extracted from the penalty function method.

Fig. 1 reports the energy management framework proposed for transmission and distribution hubs and networks. Based on this figure, there is an operator in each hub, who is aware of the data of sources, storages, and responsive loads to manage the energy of the elements inside the hub. After that, based on the upstream management by the operator of the distribution and transmission networks, it implements the optimal energy management in the hub. At first, the hub operator sends its capacity data to DSO and TSO. The hub operator simultaneously sends the amount of active generation/consumption power of the hub along with the flexibility power of the hub to the energy and flexibility market. At first, the optimal active power and flexibility of the hubs are determined by DSO and TSO, then the aforementioned markets announce the price/cost of energy and the price/income of flexibility to the hub operator. In addition, the DSO informs the TSO of its energy demand, and then the TSO informs the DSO of the optimal power level based on its goals and technical limitations. Of course, DSO and TSO announce their demand for energy and flexibility to energy and flexibility markets. Based on this information and input data from the hubs, the market establishes optimal economic conditions.

Finally, the innovations of the proposed scheme are as follows:

- Investigating the capability of flexible-renewable EHs in improving the operation and flexibility of transmission and distribution networks based on TSO-DSO coordination.
- Modeling flexibility regulation for transmission and distribution networks, and economic formulation of flexibility for flexible-renewable EHs.
- Considering CHP technology in bio-waste energy sources in EHs to increase the penetration level of RESs in the hub, as well as to add

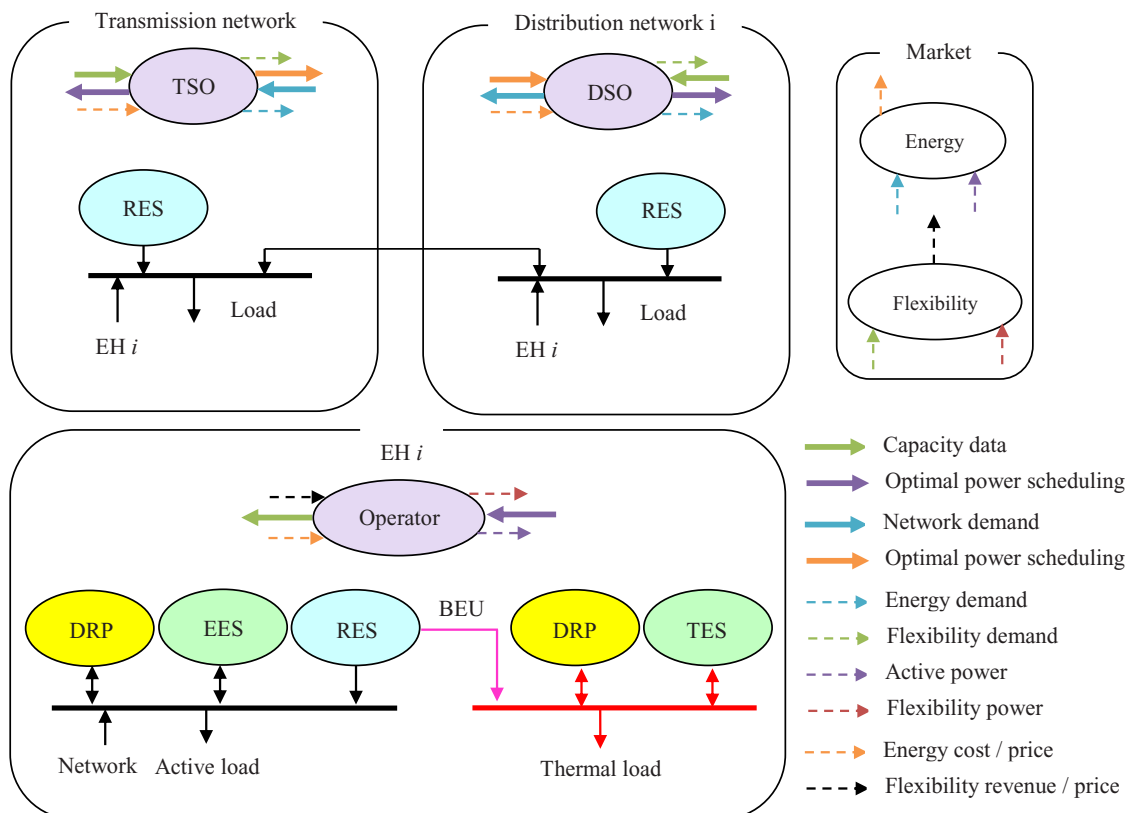


Fig. 1. Framework of the proposed scheme.

more flexibility resources such as thermal energy storage (TES) and DRP by heat demands.

- Modeling the uncertainties of load, RESs, and mobile energy storage via the unscented transformation (UT) method in the TSO-DSO coordination problem to reduce the dimension of the proposed stochastic optimization problem and reduce its computational burden.

1.5. Paper organization

The remainder of this paper is organized as follows:

The formulation of the proposed scheme based on the penalty function technique is described in Section 2. Then the uncertainty modeling based on the UT method is described in Section 3. Numerical results from different case studies are presented in Section 4, and conclusions are discussed in Section 5.

2. TSO-DSO coordination model with flexible-renewable EHs

2.1. Base formulation

In this section, the formulation of simultaneous energy management in transmission and distribution networks with flexible-renewable EHs based on the TSO-DSO coordination strategy is presented. The mentioned plan minimizes the difference between the total expected energy costs of TSO, DSO, and EHs, and the total expected flexibility revenue of EHs. The proposed optimization model is subjected to the optimal power flow and flexibility constraints of both transmission and distribution networks, as well as the EHs flexibility-operation model. In the following, the details of the formulation of the proposed scheme are presented.

A) *Objective function*: The objective function is stated in Eq. (1), which is the minimization of the difference between the total expected operating costs of networks and EHs, and the expected flexibility revenue obtained by EHs. The first line of this equation refers to the operating cost of generation units (GUs) located in the transmission network. This function is one of the goals of TSO, so it is known as TSO energy cost in this paper. In the second line of Eq. (1), the operating cost of distribution networks is modeled, which is one of the goals of DSO. It is equal to the cost of energy purchased by DSO from the TSO based on the local marginal price (LMP) in the transmission network. In the third line of Eq. (1), the operating cost of EHs is formulated. This cost is equal to the cost of purchasing EH energy from their upstream network based on LMP. In this paper, it is assumed that EHs are located both in the transmission and distribution networks. Therefore, the first (second) term of the third line of Eq. (1) refers to the purchase of EH energy in the transmission (distribution) network. In the last line of Eq. (1), the expected profit of EHs resulting from the management of network flexibility is modeled. The performance of EHs in managing flexibility is in two different upward and downward operating modes (Bagherzadeh et al., 2021). If the

EHs have more active power than the active power in the scenario corresponding to the deterministic model to regulate the flexibility of the network, the EH is in the upward mode, and the flexibility power in this mode is represented by F_{UP} . Otherwise, EH operates in the downward mode, and the flexibility power in this mode is shown by F_{DW} . Therefore, EHs can earn flexibility revenue in both upward and downward modes. In the first/second row of the last term of Eq. (1), the profit of EH flexibility in the upward/downward mode is formulated for the transmission network (first term) and distribution network (second term). Based on this formulation, the flexibility incentive price (FIP) is based on the LMPs.

In Eq. (1), the first three lines refer to the minimization of energy costs in the transmission network, distribution network, and hubs. This means that loads are supplied using sources with cheaper energy prices. Therefore, it is the same as the maximization of social welfare, following which, consumers can provide their energy consumption with the lowest energy price. To meet Eq. (1), power plants with high energy prices are committed in critical conditions such as peak load as shown in Fig. 6. The fourth line in Eq. (1) is an incentive solution for the hubs so that they operate with a high desire to adjust the flexibility of the power system with renewable resources. This section refers to improving the technical condition of the network with an economic solution.

Although different definitions for flexibility have been proposed, one of them is more used by system operators which defines flexibility as “the modification of generation injection and/or consumption patterns in reaction to an external price or activation signal in order to provide a service within the electrical system” (Bozorgavari et al., 2019). Flexibility can be provided by the supply side, network side, demand side, and energy storage systems. In other words, the power generation of renewable resources is accompanied by uncertainty. Therefore, its presence in a system leads to different results of real-time and day-ahead operations. This issue is caused by fluctuations in the power of renewable resources in the two mentioned operation modes. Hence, it is possible that the balance of generation and consumption is not established in real-time operation, which is known as a lack of flexibility. In order to improve flexibility, elements such as energy storage, non-renewable resources, and demand response need to be incorporated into the mentioned system, which are able to control their power. These elements are introduced as flexibility resources. In this case, these elements can compensate the power fluctuations of renewable resources in real-time operation compared to day-ahead operation. In these conditions, the power of the system is close to each other in the mentioned two operations, which corresponds to the high flexibility of the system. Note that, in flexibility modeling, system power differences have been investigated in different scenarios. As renewable resources produce different power in each scenario, if the flexibility resources compensate for the power fluctuations of the renewable resources in each scenario, then the optimal flexibility conditions will be achieved.

$$\begin{aligned}
 \min \quad \text{Cost} = & \overbrace{\sum_{n,h,w} \rho_w \left(a_{GU,n} + b_{GU,n} P_{GU,n,h,w} + c_{GU,n} (P_{GU,n,h,w})^2 \right)}^{\text{TSO energy cost}} \\
 & + \overbrace{\sum_{d,h,w} \rho_w \left(P_{DS,d,b=s,h,w} \sum_n K_{TD,d,n} \lambda_{TN,n,h,w} \right)}^{\text{DSOs energy cost}} \\
 & + \overbrace{\sum_{e,h,w} \rho_w \left(P_{EH,e,h,w} \left(\sum_n K_{EH,e,n} \lambda_{TN,n,h,w} + \sum_{d,b} K_{EH,d,e,b} \lambda_{DN,d,b,h,w} \right) \right)}^{\text{EHs energy cost}} \\
 & - \overbrace{\left\{ \sum_{e,h,w} \rho_w \left(F_{UP,e,h,w} \left(\sum_n K_{EH,e,n} \bar{\gamma}_{TN,n,h,w} + \sum_{d,b} K_{EH,d,e,b} \bar{\gamma}_{DN,d,b,h,w} \right) \right) + \right.}^{\text{EHs flexibility profit}} \\
 & \left. - \sum_{e,h,w} \rho_w \left(F_{DW,e,h,w} \left(\sum_n K_{EH,e,n} \bar{\gamma}_{TN,n,h,w} + \sum_{d,b} K_{EH,d,e,b} \bar{\gamma}_{DN,d,b,h,w} \right) \right) \right\}}
 \end{aligned} \quad (1)$$

B) Transmission network model: This section includes the AC optimal power flow (AC-OPF) constraints in (2)–(10), and the flexibility limit in (11), for the transmission network with RESs, EHs, and distribution networks. In the AC-OPF constraints, Eqs. (2)–(5) refer to the AC power flow model of the network, which respectively expresses the balance of active and reactive power in the network buses, and active and reactive power flow through the transmission lines (Bagheri and Jadid, 2022; Pearson et al., 2022). Constraints (6)–(7) express the modeling of the GU capability curve (Ruwaida et al., 2022), which respectively include the range of control of active and reactive power of the GU. In Eq. (8), the limit on apparent power flowing through the transmission line is presented. Constraints (9) and (10) also refer to the limitation of the voltage magnitude and angle in the transmission network. The upper limit of the voltage magnitude is to prevent the insulation vulnerability of the network equipment, the lower limit is to prevent the network outage due to severe voltage drop. The voltage angle limitation is also used to maintain the stability of the transmission network, in which the upper and lower limits of the voltage angle are at most equal to $\pi/2$ and $-\pi/2$ respectively. Constraints (8)–(10) are known as transmission network operating limitations. Finally, the flexibility limit of the transmission network is presented in Eq. (11). The power generation from renewable sources is uncertain due to its dependence on natural phenomena. Hence, the presence of these sources in the system makes the results of day-ahead and real-time operations different (Bozorgavari et al., 2019). This case is known as a lack of flexibility, as a result of which the balance of generation and consumption may not be established in real-time operation. Therefore, flexible sources (elements that can control their active power) are used in the system so that they can compensate for the power fluctuations of renewable sources in real-time operation compared to day-ahead operation (Bozorgavari et al., 2019). This is equivalent to compensating the flexibility sources of power fluctuations in different scenarios compared to the scenario corresponding to the deterministic model (here it is equal to the first scenario). In this situation, it is expected that the power of the system is close to each other in different scenarios. So, the flexibility limit for the transmission network will be in the form of (11). In the transmission network, GUs have a higher response time due to their mechanical part, so the changes in their power generations are usually slow. Considering this, in the discussion of improving the flexibility of the transmission network, the power changes of GUs in different scenarios must be low compared to the scenario corresponding to the deterministic model. Therefore, Eq. (11) is used as the flexibility limit for GUs. In this equation, the term β_F represents the flexibility tolerance, and if its value is equal to zero, the condition of 100 % flexibility is achieved. Finally, it should be noted that the price of energy in transmission network buses, i.e., λ_{TN} , is the dual variable of active power balance in (2). Also, FIP in the transmission network, i.e., $\bar{\gamma}_{TN}$ and $\underline{\gamma}_{TN}$, is the dual variable of the flexibility limit of the transmission network in (11).

$$P_{GU\ n,h,w} + P_{R\ n,h,w} + \sum_j K_{TL\ nj} P_{TL\ nj,h,w} + \sum_e K_{EH\ e,n} P_{EH\ e,h,w} - \sum_d K_{TD\ dn} P_{DS\ d,b=s,h,w} = \lambda_{TN\ n,h,w} \quad \forall n, h, w \quad (2)$$

$$Q_{GU\ n,h,w} + \sum_j K_{TL\ nj} Q_{TL\ nj,h,w} + \sum_d K_{TD\ dn} Q_{DS\ d,b=s,h,w} = Q_{D\ n,h,w} \quad \forall n, h, w \quad (3)$$

$$P_{TL\ nj,h,w} = G_{TL\ nj} (V_{n,h,w})^2 - V_{n,h,w} V_{j,h,w} (G_{TL\ nj} \cos(\delta_{n,h,w} - \delta_{j,h,w}) + B_{TL\ nj} \sin(\delta_{n,h,w} - \delta_{j,h,w})) \quad (4)$$

$$Q_{TL\ nj,h,w} = -B_{TL\ nj} (V_{n,h,w})^2 + V_{n,h,w} V_{j,h,w} (B_{TL\ nj} \cos(\delta_{n,h,w} - \delta_{j,h,w}) - G_{TL\ nj} \sin(\delta_{n,h,w} - \delta_{j,h,w})) \quad (5)$$

$$\underline{P}_{GU\ n} \leq P_{GU\ n,h,w} \leq \bar{P}_{GU\ n} \quad \forall n, h, w \quad (6)$$

$$\underline{Q}_{GU\ n} \leq Q_{GU\ n,h,w} \leq \bar{Q}_{GU\ n} \quad \forall n, h, w \quad (7)$$

$$\sqrt{(P_{TL\ nj,h,w})^2 + (Q_{TL\ nj,h,w})^2} \leq \bar{S}_{TL\ nj} \quad \forall n, j, h, w \quad (8)$$

$$\underline{V}_n \leq V_{n,h,w} \leq \bar{V}_n \quad \forall n, h, w \quad (9)$$

$$-\bar{\delta}_n \leq \delta_{n,h,w} \leq \bar{\delta}_n \quad \forall n, t, w \quad (10)$$

$$-\beta_F \leq P_{GU\ n,h,w} - P_{GU\ n,h,w=1} \leq \beta_F : \underline{\gamma}_{TN\ n,h,w}, \bar{\gamma}_{TN\ n,h,w} \quad \forall n, h, w \quad (11)$$

C) Distribution networks model: The distribution network model includes the AC-PF constraints (12)–(18), and the flexibility constraint (19), in this network. The formulation of AC power flow in the distribution network is according to (12)–(15), which refers to the balance of active and reactive power in the network buses, and active and reactive power flow through the distribution line. The operating limit of this network is modeled in (16)–(18). The limitation of the distribution network operation includes the limitation of the apparent power flow through the distribution line and the distribution substation, (16) and (17), and the limitation of the voltage magnitude of the buses, (18) (Fatemi et al., 2023). It is assumed that the distribution network is connected to the transmission network through the distribution substation located in the slack bus (s). Therefore, P_{DS} and Q_{DS} variables only have values for $b = s$ and are equal to zero for other buses. Flexibility constraint is modeled in (19). From the TSO point of view, the power of the distribution network is equal to the power passing through the distribution substation. Hence, the flexibility limit for the distribution substation is written based on Eq. (19). The energy price for distribution network buses, λ_{DN} , corresponds to the dual variable of active power balance in buses, (12). FIP in this network, $\bar{\gamma}_{DN}$ and $\underline{\gamma}_{DN}$, is equal to the dual variable of the flexibility limit of the distribution network, (19).

$$P_{DS\ d,b,h,w} + P_{R\ d,b,h,w} + \sum_j K_{DL\ db,j} P_{DL\ d,b,j,h,w} + \sum_e K_{EH\ e,b} P_{EH\ e,h,w} = P_{D\ d,b,h,w} : \lambda_{DN\ d,b,h,w} \quad \forall d, b, h, w \quad (12)$$

$$Q_{DS\ d,b,h,w} + \sum_j K_{DL\ db,j} Q_{DL\ d,b,j,h,w} = Q_{D\ d,b,h,w} \quad \forall d, b, h, w \quad (13)$$

$$P_{DL\ d,b,j,h,w} = G_{DL\ db,j} (V_{d,b,h,w})^2 - V_{d,b,h,w} V_{d,j,h,w} \left(G_{DL\ db,j} \cos(\delta_{d,b,h,w} - \delta_{d,j,h,w}) + B_{DL\ db,j} \sin(\delta_{d,b,h,w} - \delta_{d,j,h,w}) \right) \quad \forall d, b, j, h, w \quad (14)$$

$$Q_{DL\ d,b,j,h,w} = -B_{DL\ db,j} (V_{d,b,h,w})^2 + V_{d,b,h,w} V_{d,j,h,w} \left(B_{DL\ db,j} \cos(\delta_{d,b,h,w} - \delta_{d,j,h,w}) - G_{DL\ db,j} \sin(\delta_{d,b,h,w} - \delta_{d,j,h,w}) \right) \quad \forall d, b, j, h, w \quad (15)$$

$$\sqrt{(P_{DL\ d,b,j,h,w})^2 + (Q_{DL\ d,b,j,h,w})^2} \leq \bar{S}_{DL\ d,b,j} \quad \forall d, b, j, h, w \quad (16)$$

$$\sqrt{(P_{DS\ d,b,h,w})^2 + (Q_{DS\ d,b,h,w})^2} \leq \bar{S}_{DS\ d,b} \quad \forall d, b = s, h, w \quad (17)$$

$$\underline{V}_{d,b} \leq V_{d,b,h,w} \leq \bar{V}_{d,b} \quad \forall d, b, h, w \quad (18)$$

$$-\beta_F \leq P_{DS\ d,b,h,w} - P_{DS\ d,b,h,w=1} \leq \beta_F : \gamma_{DN\ d,b,h,w}, \bar{\gamma}_{DN\ d,b,h,w} \quad \forall d, b, h, w \quad (19)$$

D) EHs model: In Eq. (20), the active power balance in the EH is modeled. Based on this equation, the excess generation (consumption) power is injected (received) into (from) the network as a P_{EH} variable. In Eq. (21), the thermal power balance in EH is considered. Based on this relationship, sources, storage, and thermal DRP should supply the thermal load in EH. In this paper, it is assumed that the BEU is equipped with CHP technology, so it can produce thermal power equal to $\vartheta \times P_R$ (Homayoun et al., 2021). In Eqs. (22)–(24), the formulation of electrical energy storage (EES) is presented. The constraints on the EES discharge and charge rates are given in constraints (22) and (23), respectively. The limit of energy stored in EES is expressed in (24) (Bozorgavari et al., 2019). It is worth noting that the formulation of the mobile EES (aggregation of EVs) is the same as (22)–(24), except for $0 \leq \dot{E}_{ES} - \bar{E}_{ES} + \sum_{t=1}^h \left(\eta_{CH} P_{CH\ e,t,w} - \frac{1}{\eta_{DC}} P_{DC\ e,t,w} \right)$ will substitute Eq. (24). Also, the parameter $\bar{P}_{DC}, \bar{P}_{CH}, \bar{E}_{ES}$, and $\dot{E}_{ES}$ change according to time (h) and scenario (w) because the number and type of EVs connected to the parking lot are different at every moment and every scenario. The parameters $\bar{P}_{DC}$ and $\bar{P}_{CH}$ at each moment and scenario are respectively equal to the total discharge and charge rates of all EVs connected to the parking lot. The parameter $\dot{E}_{ES}(\bar{E}_{ES})$ at each moment and scenario is equal to the sum of the initial energy (energy required for travel) of all newly connected (disconnected) EVs to (from) the parking lot (Bagherzadeh and Kamwa, 2023; Bagherzadeh et al., 2019). The energy consumption required for the travel of an EV is equal to the product of its battery capacity (BC) and state of charge (SOC) (Nasir et al., 2023; Shayanfar et al., 2023). The formulation of the thermal energy storage (TES) is presented in the constraints (25)–(27), which is the same as the EES model, except for the thermal charge and discharge variables, H_{CH} and H_{DC} , are used in (25)–(27) (AkbariZadeh et al., 2021; Homayoun et al., 2021). The DRP model for electrical and thermal consumers is presented in (28)–(29) and (30)–(31), respectively (Mansouri et al., 2022). This DRP is based on energy prices, i.e., λ_{TN} and λ_{DN} so that when the price of energy is low (high), consumers increase (decrease) their energy consumption. This reduces the operating cost of EHs based on the third line of Eq. (1). Based on this, constraints (28) and (30) respectively express the range of control of active and thermal power of consumers in the DRP. Constraints (29) and (31) also guarantee that the total energy reduced in high energy price hours is supplied from EHs in low energy price hours (Mansouri et al., 2022). Finally, the flexibility model of EHs is shown in (32)–(33). As stated in subsection 2. A, there are two operation modes, namely, upward and downward, for EHs in regulating the flexibility of networks. In both modes, the flexibility power of EH is equal to the difference between the active power of EH in one scenario and that of the scenario corresponding to the deterministic model (first scenario) (Bozorgavari et al., 2019). If the amount of flexibility power is a positive (negative) value, EH operates in the upward (downward) mode. This is formulated in (32), and the limitation of flexibility power in the two mentioned modes is proportional to constraint (33).

$$P_{EH\ e,h,w} = P_{R\ e,h,w} + (P_{DC\ e,h,w} - P_{CH\ e,h,w}) + P_{DR\ e,h,w} - P_{D\ e,h,w} \quad \forall e, h, w \quad (20)$$

$$\begin{aligned} \vartheta_e P_{R\ e,h,w} + (H_{DC\ e,h,w} - H_{CH\ e,h,w}) + H_{DR\ e,h,w} \\ = H_{D\ e,h,w} \quad \forall e, h, w \\ = \frac{1 - \eta_t - \eta_h}{\eta_t} \eta_h \quad \text{for BEU, } \vartheta = 0 \quad \text{for WT and PV} \end{aligned} \quad (21)$$

$$0 \leq P_{DC\ e,h,w} \leq \bar{P}_{DC\ e} \quad \forall e, h, w \quad (22)$$

$$0 \leq P_{CH\ e,h,w} \leq \bar{P}_{CH\ e} \quad \forall e, h, w \quad (23)$$

$$\underline{E}_{ES} \leq \dot{E}_{ES} + \sum_{t=1}^h \left(\eta_{CH} P_{CH\ e,t,w} - \frac{1}{\eta_{DC}} P_{DC\ e,t,w} \right) \leq \bar{E}_{ES} \quad \forall e, h, w \quad (24)$$

$$0 \leq H_{DC\ e,h,w} \leq \bar{H}_{DC\ e} \quad \forall e, h, w \quad (25)$$

$$0 \leq H_{CH\ e,h,w} \leq \bar{H}_{CH\ e} \quad \forall e, h, w \quad (26)$$

$$\underline{E}_{TS} \leq \dot{E}_{TS} + \sum_{t=1}^h \left(\eta_{CH} H_{CH\ e,t,w} - \frac{1}{\eta_{DC}} H_{DC\ e,t,w} \right) \leq \bar{E}_{TS} \quad \forall e, h, w \quad (27)$$

$$-\sigma_e P_{D\ e,h,w} \leq P_{DR\ e,h,w} \leq \sigma_e P_{D\ e,h,w} \quad \forall e, h, w \quad (28)$$

$$\sum_h P_{DR\ e,h,w} = 0 \quad \forall e, w \quad (29)$$

$$-\sigma_e H_{D\ e,h,w} \leq H_{DR\ e,h,w} \leq \sigma_e H_{D\ e,h,w} \quad \forall e, h, w \quad (30)$$

$$\sum_h H_{DR\ e,h,w} = 0 \quad \forall e, w \quad (31)$$

$$F_{UP\ e,h,w} - F_{DW\ e,h,w} = P_{EH\ e,h,w} - P_{EH\ e,h,w=1} \quad \forall e, h, w \quad (32)$$

$$F_{UP\ e,h,w}, F_{DW\ e,h,w} \geq 0 \quad \forall e, h, w \quad (33)$$

2.2. Integrated TSO-DSO Coordinated energy market model

In the formulation of the proposed model for the joint energy and flexibility market in (1)–(33), it is necessary to simultaneously determine the main variables of the problem such as active and reactive power, and dual variables such as energy price and FIP. For this purpose, the penalty function technique (Jokar et al., 2023; Yang et al., 2021) is used in this paper. In this approach, the objective function is equal to the cost function defined in (1) and the sum of the penalty functions. The added terms which are referred to as the penalty functions are also equal to $\lambda \times (b - a)$ and $\gamma \times \max(0, a - b)$ for a constraint $a = b$ and a constraint $a \leq b$, respectively. Also, $\lambda \in (-\infty, +\infty)$ and $\gamma \geq 0$ represent dual variables of equality and inequality constraints (Jokar et al., 2023; Yang et al., 2021). Therefore, according to the topic, the objective function in the integrated TSO-DSO model for the proposed scheme is in the form of (34). It is equal to the sum of the objective function (1), cost, and the sum of the penalty functions of the constraints (2), (11), (12), and (19). In this problem, the same constraints of Eq. (35) include the base formulation constraints in the previous section, with the difference that Eqs. (2), (11), (12), and (19) are removed because these equations appeared as a penalty function in the objective function.

$$\begin{aligned}
\min \quad & \text{Cost} + \\
& \sum_{n,h,w} \lambda_{TN,n,h,w} (\text{right side of (2)} - \text{left side of (2)}) + \\
& \sum_{d,b,h,w} \lambda_{DN,d,b,h,w} (\text{right side of (12)} - \text{left side of (12)}) + \\
& \sum_{n,h,w} \left(\frac{\gamma_{TN,n,h,w}}{\bar{\gamma}_{TN,n,h,w}} \max(0, \beta_F - (P_{GU,n,h,w} - P_{GU,n,h,w=1})) + \right) + \\
& \sum_{d,b,h,w} \left(\frac{\gamma_{DN,d,b,h,w}}{\bar{\gamma}_{DN,d,b,h,w}} \max(0, P_{DS,d,b,h,w} - P_{DS,d,b,h,w=1} - \beta_F) \right)
\end{aligned} \quad (34)$$

Subject to:

$$\text{Constraints (3)-(10), (13)-(18), (20)-(33)} \quad (35)$$

Due to the non-fulfillment of constraints (2), (11), (12), and (19) in the above formulation and the high value of dual variables, the unbounded state may be identified.

The optimal point is obtained when the constraints (2), (11), (12), and (19) are satisfied, by repeating the different values for dual variables and then evaluating the condition of the constraint.

As a starting point, an initial value for the dual variables λ and γ is considered in Eq. (34). The Eq. (34)-(35) are then solved. The deviation of constraints (2), (11), (12), and (19) can be investigated by solving this problem. In other words, in these conditions, the value of ε_{DT} (active power balance deviation in the transmission network), ε_{DD} (active power balance deviation in the distribution network), ε_{FGU} (production unit flexibility constraint deviation in upward mode), ε_{FGD} (distribution deviation of production units in downward mode), ε_{FDU} (distribution network flexibility limitation in upward mode), ε_{FDD} (distribution network flexibility limitation deviation in downward mode) are calculated by Eqs. (36)-(41).

If the values of ε_{DT} and ε_{DD} have a positive (negative) value, the value of the dual variable λ must be increased (decreased) with a specific step in the next stage. Also, if the value of ε_{FGU} , ε_{FGD} , ε_{FDU} , and ε_{FDD} is negative, the value of the dual variable γ should be increased by a certain amount in the next stage.

The Eq. (34)-(35) as well as the values of ε_{DT} , ε_{DD} , ε_{FGU} , ε_{FGD} , ε_{FDU} , and ε_{FDD} are calculated again with the new values of the dual variables λ and γ . In the new stage, if the value of these variables is very close to zero, the optimal solution has been extracted. Otherwise, the values of the dual variables should be updated again based on the described explanations.

$$\begin{aligned}
\varepsilon_{DT,n,h,w} = & P_{D,n,h,w} - \left(P_{GU,n,h,w} + P_{R,n,h,w} + \sum_j K_{TL,n,j} P_{TL,n,j,h,w} + \sum_e K_{EH,e,n} P_{EH,e,h,w} - \sum_d K_{TD,d,n} P_{DS,d,b=s,h,w} \right) \\
& \forall n, h, w
\end{aligned} \quad (36)$$

$$\begin{aligned}
\varepsilon_{DD,d,b,h,w} = & P_{D,d,b,h,w} - \left(P_{DS,d,b,h,w} + P_{R,d,b,h,w} + \sum_j K_{DL,d,b,j} P_{DL,d,b,j,h,w} + \sum_e K_{EH,d,e,b} P_{EH,e,h,w} \right) \\
& \forall d, b, h, w
\end{aligned} \quad (37)$$

$$\varepsilon_{FGU,n,h,w} = \min(0, \beta_F - (P_{GU,n,h,w} - P_{GU,n,h,w=1})) \quad \forall n, h, w \quad (38)$$

$$\varepsilon_{FGD,n,h,w} = \min(0, -\beta_F - (P_{GU,n,h,w} - P_{GU,n,h,w=1})) \quad \forall n, h, w \quad (39)$$

$$\varepsilon_{FDU,d,b,h,w} = \min(0, \beta_F - (P_{DS,d,b,h,w} - P_{DS,d,b,h,w=1})) \quad \forall d, b, h, w \quad (40)$$

$$\varepsilon_{FDD,d,b,h,w} = \min(0, -\beta_F - (P_{DS,d,b,h,w} - P_{DS,d,b,h,w=1})) \quad \forall d, b, h, w \quad (41)$$

3. Uncertainty modeling based on the UT method

In the proposed model, the load (P_D , Q_D , and H_D), renewable power (P_R), and parameters of EVs ($\bar{P}_{DC}$, $\bar{P}_{CH}$, $\bar{E}_{ES}$, and $\dot{E}_{ES}$) are uncertain quantities. This section addresses the third research gap given in subsection 1.3. Also, prediction error related to the aforementioned uncertainties is suitably handled by using UT-based stochastic optimization (Kazemi et al., 2022). Thanks to its benefits, including fast calculation and short convergence duration, the UT is conceived as one of the efficient tools to deal with and model uncertain parameters. There are no specific assumptions in this method and it can easily deal with nonlinear behavior of uncertain parameters (Kazemi et al., 2022) and provides an accurate estimate for the probability distribution function. For these reasons, it is adopted in the current study to incorporate the uncertainties in the proposed model. Parameter m is the dimension of the vector (U) that includes input uncertainties. A significant point to note is that for modeling uncertainty, the UT method will generate $2m+1$ scenarios containing m uncertain variables (Dabbaghjamesh et al., 2018).

The problem under discussion, shown by $y = f(x)$, is uncertain, nonlinear, and stochastic. Parameter y is a vector that contains uncertain outputs and x is a vector of uncertain inputs. The mean vector and covariance matrix of input parameters x are represented by μ_x and σ_x . Variance and covariance of uncertainties can be obtained from diagonal and off-diagonal entries of σ_x , respectively. To find μ_x and σ_x , of the output, the UT method follows several steps as described below (Dabbaghjamesh et al., 2018):

- Step 1: Select $2m+1$ samples from the input uncertain data:

$$x_0 = \mu_x \quad (42)$$

$$x_\varpi = \mu_x + \sqrt{\frac{m}{1-W^0}} \sigma_x \quad \forall \varpi = 1, 2, \dots, m \quad (43)$$

$$x_\varpi = \mu_x - \sqrt{\frac{m}{1-W^0}} \sigma_x \quad \forall \varpi = 1, 2, \dots, m \quad (44)$$

Where W^0 is the weight of μ_x .

- Step 2: Investigate the weighting factor of samples:

$$W^0 = W^0 \quad (45)$$

$$W_\varpi = \frac{1-W^0}{2m} \quad \forall \varpi = 1, 2, \dots, m \quad (46)$$

$$W_{\varpi+n} = \frac{1-W^0}{2m} \quad \forall \varpi + m = m+1, m+2, \dots, 2m \quad (47)$$

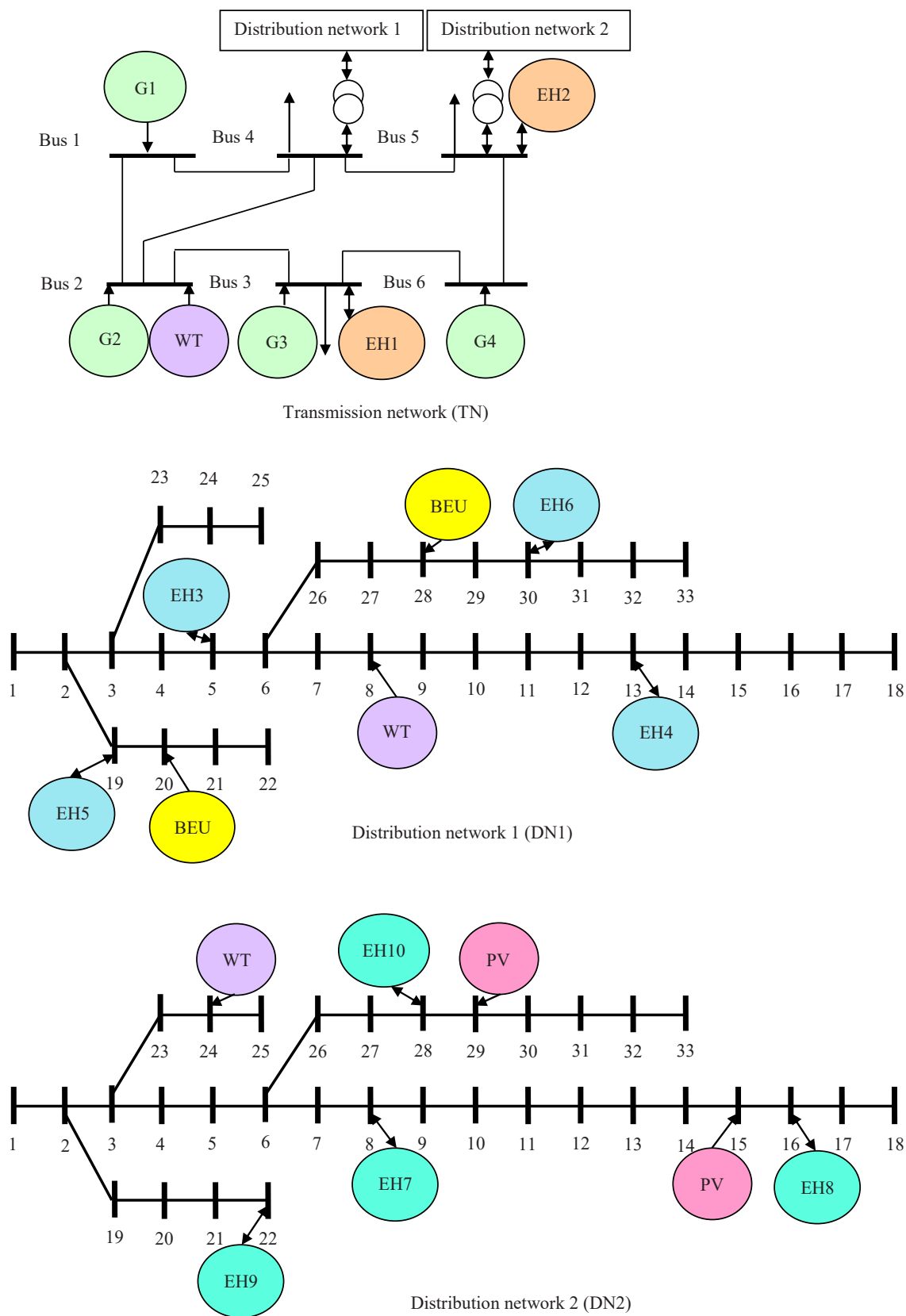


Fig. 2. IEEE 6-bus transmission network (Hamidpour et al#, 2021) including sources, EHs, and IEEE 33-bus radial distribution network (Babu et al#, 2009).

Table 2

Data of RESs and peak demand within EHs.

EH	Capacity (MW) of			Load (MW)	
	WT	BEU	PV	P_D	H_D
1	3	0	0	2	0
2	0	3	0	1.5	1
3	0.6	0	0	0.2	0
4	0.3	0	0.3	0.2	0
5	0	0.8	0	0.2	0.2
6	0	0.3	0.3	0.2	0.1
7	0	0.6	0.2	0.2	0.2
8	0	0	0.4	0.2	0
9	0.6	0	0	0.2	0
10	0.3	0.3	0	0.2	0.1

$$\sum_{m=0}^{2m} W_m = 1 \quad (48)$$

- *Step 3:* Give $2m+1$ points to the nonlinear function to find a set of transformed output samples as:

$$y_m = f(x_m) \quad (49)$$

- *Step 4:* Investigate σ_y and μ_y of the output variable ν :

$$\mu_y = \sum_{m=0}^{2m} W_m \nu_m \quad (50)$$

$$\sigma_y = \sum_{m=0}^{2m} W_m (\nu_m - \mu_y) (\nu_m - \mu_y)^T \quad (51)$$

The proposed scheme is based on the optimization model. This model includes an objective function, where it contains a min term. There are different constraints which are equality and inequality forms. To apply the optimization model, the network needs to be smart devices. Smart devices are based on intelligent algorithms and communication tools.

4. Numerical results

In general, the proposed scheme has no limitations of being applied to various data of networks, sources, storage devices, and responsive loads. In this section, the proposed model is applied to IEEE 6-bus and 30-bus transmission networks. These networks also are connected to the IEEE 33-bus distribution network as their downstream low-voltage (LV) grid. The proposed optimization model is implemented in the GAMS software environment (Boisvert et al., 1985), and the IPOPT solver (Boisvert et al., 1985) is adopted to solve it.

4.1. IEEE 6-bus transmission network

A) *Data:* As shown in Fig. 2, in this case, the IEEE 6-bus transmission network (TN) (Hamidpour et al., 2021) is connected to two IEEE 33-bus radial distribution networks (DN) (Babu et al., 2009) at buses 4 and 5. The data of transmission lines and generation units, i.e., G1–G4, are extracted from (Hamidpour et al., 2021). The transmission network alone has a peak load of 22 MW, which was distributed in buses 3–5. Bus 3 contains 40 % of the total load, while buses 4 and 5 contain 30 % of the total demand. The power factor of demand in the transmission buses is assumed to be 0.92 lag. The data of distribution feeder, substations, and peak loads for the distribution network are available in (Babu et al., 2009). The base power is 1 MVA, and the base voltage of the transmission and distribution network is 63 kV and 12.66 kV, respectively. The voltage magnitude in the transmission and distribution networks should be within the permitted range of [0.95 1.05] and [0.9 1.05] p.u.,

respectively. In this paper, to maintain angular stability, the voltage angle in the transmission network should be between $\pi/6$ and $-\pi/6$ (Hamidpour et al., 2021). The transmission network includes a wind farm with a capacity of 4 MW, whose location is shown in Fig. 2. The capacity of the wind turbine in DN1 and DN2 is 0.8 MW and 1 MW, respectively. In DN1, two BEUs each with 0.5 MW capacity are used, and in DN2, two PVs each with 0.3 MW capacity are employed. The location of these RESs is also shown in Fig. 2. In the proposed scheme, to achieve high flexibility in transmission and distribution networks, the flexibility tolerance is set at 0.05 p.u.

Concerning EHs, their locations in the transmission and distribution networks are presented in Fig. 2. The capacity of WT, PV, and BEU in EHs is according to the data in Table 2. Turbine, losses, and heat efficiencies are assumed to be the same for all BEUs, which are 40 %, 9 %, and 40 %, respectively (Homayoun et al., 2021). Table 2 reports the amount of peak active and thermal load for each EH. In all EHs, electrical and thermal consumers participate in the DRP with a participation rate of 40 %. Moreover, the EHs located in the distribution network, namely EH3–EH10, each have a parking lot with a capacity of 50 EVs. The characteristics of an EV such as SOC, battery capacity, charging and discharging rates, and charging and discharging efficiency are specified in (Bagherzadeh and Kamwa, 2023). The hourly load data is equal to the product of the peak load and the load factor, and the expected daily curve of the electric and thermal load factor is shown in Fig. 3(a). The active power generation by each RES is the product of its capacity and the rate of generated power. The expected daily curve of the power generation rate of WT, PV, and BEU is given in Fig. 3(b) (Jokar et al., 2023). Also, at each hour, the number of EVs in the parking lot equals the product of the total number of EVs and EVs penetration rate. The expected daily curve of the EV penetration rate is depicted in Fig. 3(c) (Jokar et al., 2023). The EHs located in the transmission network, namely EH1–EH2, have EES with 1, 0.5 MWh minimum energy, and 0.5 MWh initial energy. These data for EES in EHs 3–10 are 90 %, 0.6 MWh, 1.5 MWh, 0.15 MWh, and 0.15 MWh, respectively. The data of the TESs is the same as EESs, except that their charge and discharge efficiency is 79 %. Moreover, TESs are present in hubs that have BEU and thermal load. Finally, the uncertainty standard deviation is set at 10 %.

B) *Operating status of EHs and networks:* Fig. 4 shows the expected curve of active and thermal power of sources, storages, and responsive loads along with EHs for the TN, DN1, and DN2. Based on this figure, the active power obtained from RESs in EHs every hour is equal to the sum of the active power of PVs, WTs, and BEUs. Also, as Table 2 shows, the capacity of RESs in EHs is higher than the peak load of EHs, so the power generation by the RESs in each EH is higher than its passive demand in most hours as shown in Fig. 4. The power change of storages and responsive loads in EHs is proportional to the daily load curve. During the off-peak period, from 1:00–7:00 and 23:00–00:00, according to Fig. 3(a), the mentioned elements are in charging mode and receive power from EHs. During peak hours, such as between 16:00 and 22:00, storages and responsive loads are in discharge mode and inject energy to EHs. This is because during off-peak hours, the amount of load is low, so it is expected that power plants with lower fuel prices (such as G1 and G2, which have the lowest fuel prices (Hamidpour et al., 2021)) play a role in supplying energy. However, during peak hours, power plants with higher fuel prices also start up, so the price of energy is low in off-peak hours and high in peak hours. Therefore, to reduce the energy cost of EHs and networks as given in Eq. (1), the storage devices and responsive loads operate in the charging (discharging) mode during off-peak (peak) hours because this period corresponds to low (high) energy prices. This mode of operation leads to the reduction of energy costs in EHs and networks. Another factor of this type of operation of storage and responsive loads is the difference between the power generation by the RESs and the consumption power of passive loads in EHs. So, in the hours that the power of RESs is greater than the load, the mentioned elements are in charging mode. As the difference between RESs power and passive load decreases, the storage and responsive loads

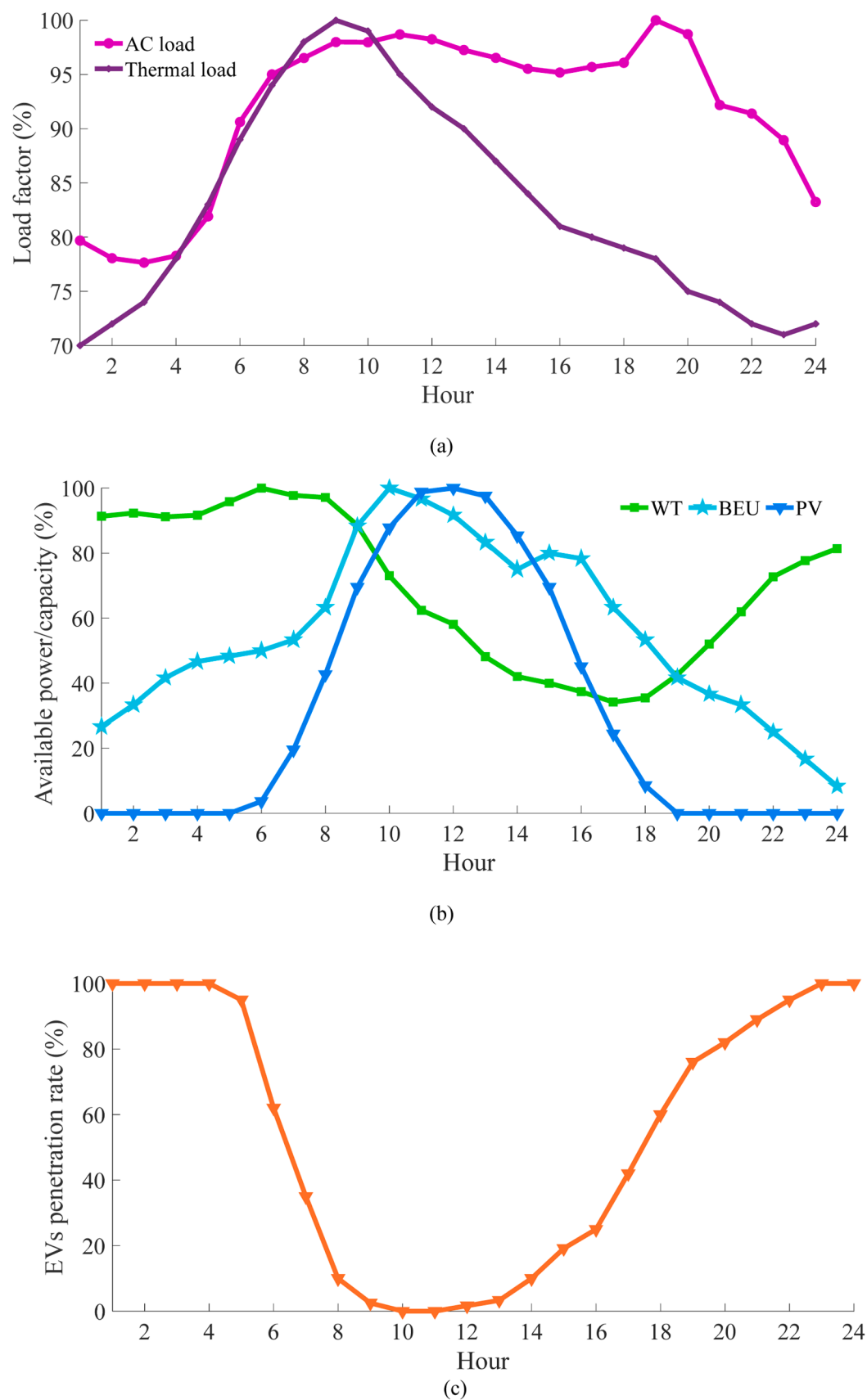


Fig. 3. Expected daily curve of, a) electricity and thermal load factor (Hamidpour et al., 2021), b) generation power rate of WT (Jokar et al., 2023), BEU (Jokar et al., 2023), and PV (Homayoun et al., 2021), and c) EVs penetration rate (Jokar et al., 2023).

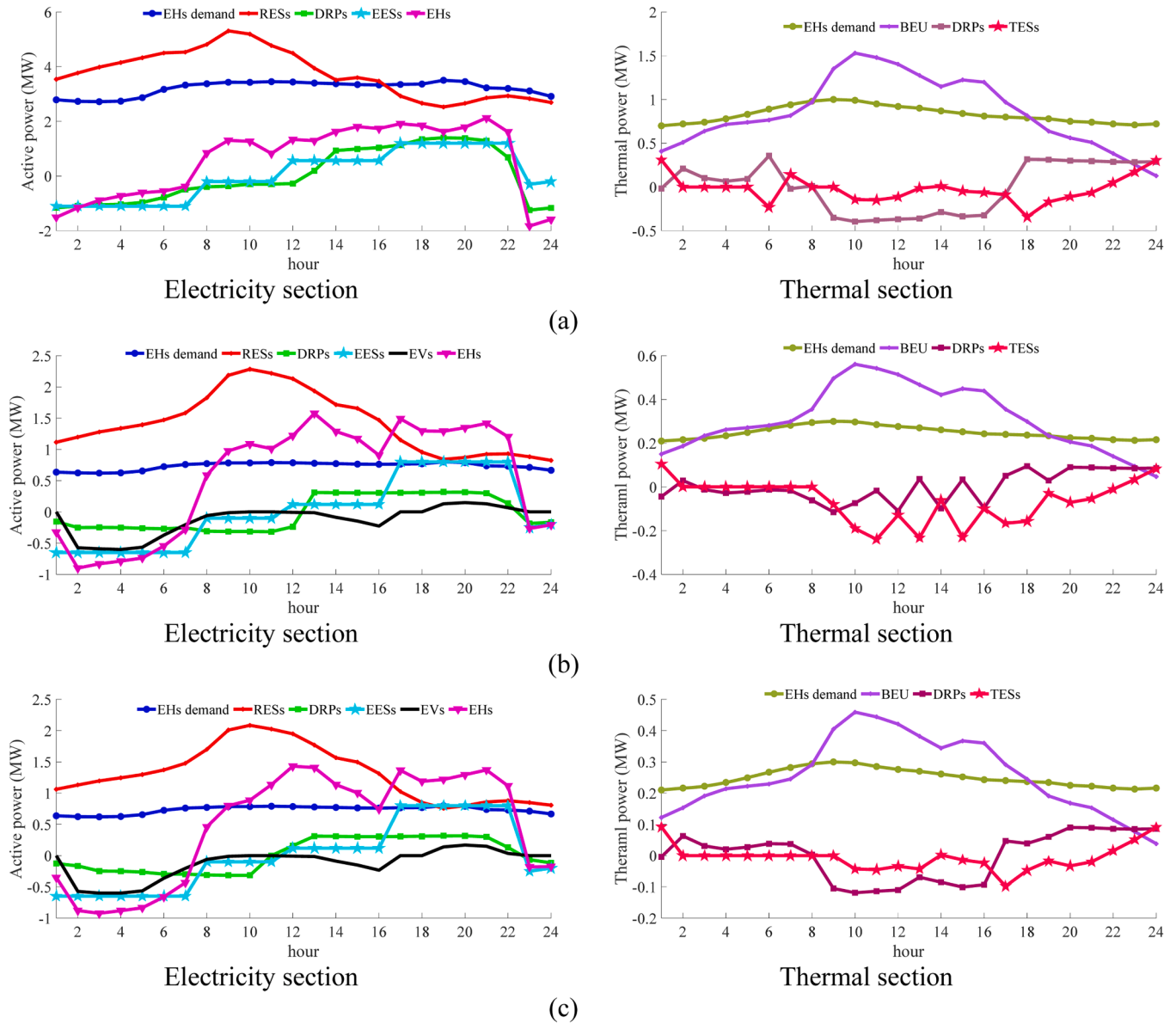


Fig. 4. Expected daily active and thermal power curve of EHs and their devices in, a) transmission network, b) distribution network 1, c) distribution network 2.

shift towards power injection move. In terms of EVs performance, they receive high energy from EHs in the hours 2:00–9:00. This energy corresponds to their energy consumption for travel. They received this amount of energy from EHs during the mentioned hours because the energy price is low, and the RESs power is more than the passive load power. Moreover, they are in charging mode from 12:00–15:00 and are in discharge mode from 19:00–22:00. In other words, they receive energy from EHs at 12:00–15:00 until they can inject energy into EHs at 19:00–22:00 to reduce energy costs. Finally, according to this type of operation of sources, storages, and responsive loads, according to Fig. 4, EHs can produce electrical energy most of the time. They are in the role of consumers at 1:00–7:00 and 23:00–00:00 and in other hours they produce electrical energy. Regarding the operation of EH elements in the thermal section, based on Fig. 4, it can be seen that the operation of sources, storages, and responsive loads is such that the balance of generation and consumption in EHs is maintained during all operating hours. So, in the hours when the thermal power of BEU is more (less) than the thermal power consumption, the total power of DRPs and TESs is in consumption (generation) mode. It should be noted that based on Eq. (21), only BEU plays a role in thermal power production.

In Fig. 5, the expected daily curve of the active power of RESs and the distribution substation for three study cases of the network without RESs and EHs, network without EHs, and network with RESs and EHs for two distribution networks DN1 and DN2 are drawn. Based on this, the active power of RESs in the distribution network for every hour is equal to the total active power of photovoltaics, BEUs, and WTs. The only presence of RESs in the distribution network has caused the active power passing through the distribution substation to decrease compared to the power flow studies of the networks (network without RESs and EHs) in all operating hours. This means that DSOs provide a specific part of the energy consumption of passive loads by RESs in the case study of the network without EHs, so their energy demand from TSOs decreases at all hours. However, in the case of studies where EHs are added to the network (i.e., network with EHs and RESs), based on Fig. 5, the active power flowing through the distribution substation for the hours 1:00–7:00 and 23:00–00:00 has increased compared to the case without EHs, but it decreases in other hours. This is because, according to Fig. 4, EHs are in the mode of consuming electric energy in the mentioned hours, and in other hours they play the role of producing electrical energy. Therefore, the presence of EHs in the distribution network

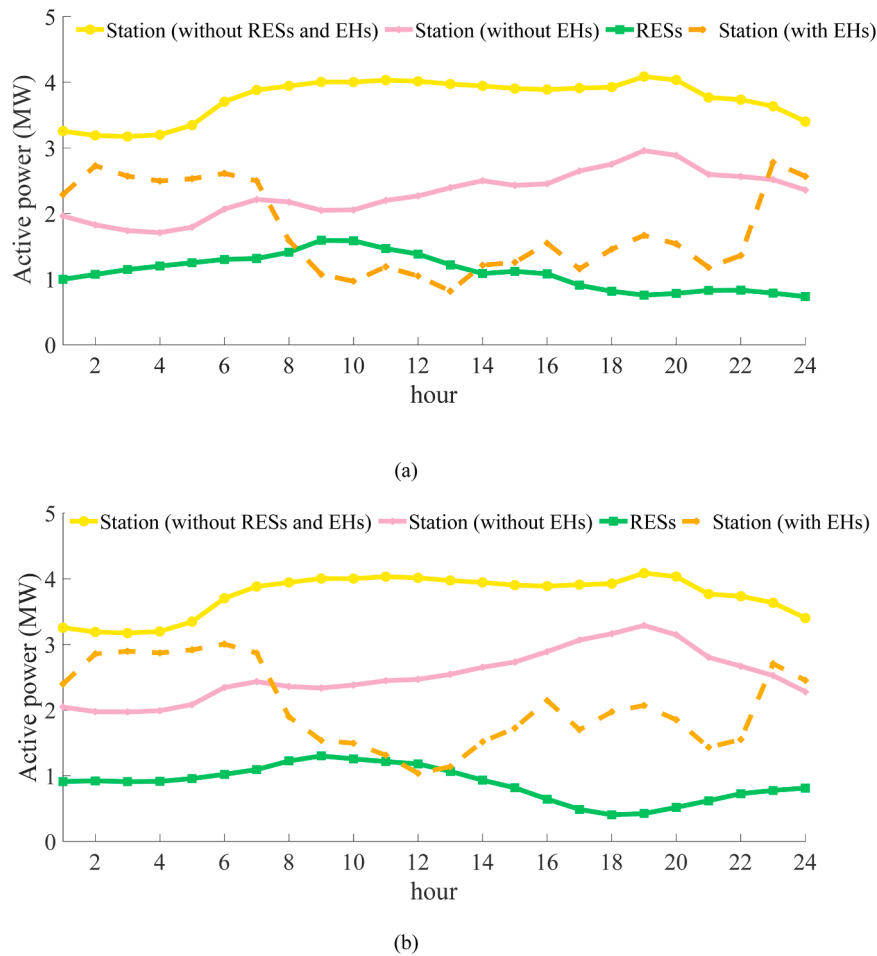


Fig. 5. Expected daily active power curve of the station and renewable sources in the distribution network, a) 1, b) 2.

increases the active power demand of DSO from TSO in off-peak hours, 1:00–7:00 and 23:00–00:00. Yet, in other operating hours, EH leads to a decrease in the active power demand of DSO from TSO. The expected daily curve of the active power generation by power plants and the wind farm is depicted in Fig. 6 for two cases with/without the presence of EHs. In the case without EHs, based on Fig. 6(a), G1 and G2, which have lower fuel costs (Hamidpour et al., 2021), produce higher active power during all operating hours. But G4, which has a high fuel price, starts up in fewer hours. Its startup occurs when the amount of load is the highest and the power generation by the wind farm is the lowest. With the presence of EHs in distribution and transmission networks, the amount of active power generation by G2 and G3 increases in the hours 1:00–7:00 and 23:00–00:00 compared to the case without EHs, but in other hours their power generation has decreased. Also, the presence of EHs has led the G4, which has a high fuel price, to operate in fewer hours. Therefore, the presence of EHs in distribution and transmission networks and their proper energy management will reduce energy costs.

According to (Hamidpour et al., 2021), G1 has a lower fuel cost than G2, G3 and G4. Therefore, to minimize the cost of energy, as in the first line of Eq. (1), this generation unit is the priority of energy supply. Since the level of energy demand from power plants in two cases, i.e., with and without EH, is more than the 10 MW capacity. So, G1 always produces power equal to its maximum capacity in two studied cases.

Table 3 reports the values of operation indicators such as energy loss and maximum voltage deviation for distribution networks DN1 and DN2 and transmission network (TN) for the study cases without RESs and EHs, without EHs, and with RESs and EHs. According to this table, the transmission and distribution networks have energy losses of 17.6 MWh and 2.97 MWh respectively in power flow studies (network without

RESs and EHs). Their maximum voltage deviation is also equal to 0.012 p.u. and 0.087 p.u. Nonetheless, the presence of RESs has been able to reduce energy losses in the transmission network by about 11.4 % ((17.6–15.6)/17.6) compared to power flow studies. The loss reduction in DN1 and DN2 is about 8.1 % and 6.4 %, respectively. Finally, RESs can reduce the total energy losses in transmission and distribution networks by about 10.3 % compared to power flow studies. The maximum voltage deviation in TN, DN1, and DN2 networks with RESs decreases by about 8.3 %, 9.2 %, and 5.7 %, respectively, compared to the case without EHs and RESs. Further, with the simultaneous addition of EHs and RESs to the distribution and transmission networks, it is more desirable to improve the operation indicators. The total energy losses and the maximum voltage deviation in TN, DN1, and DN2 are reduced by 22.2 %, 25 %, 33.3 %, and 31 %, respectively, compared to power flow studies.

C) *Investigating the performance of EHs in adjusting the flexibility of distribution and transmission networks:* The daily expected flexibility power curve of EHs in transmission and distribution networks for two upward and downward modes is drawn in Fig. 7. By comparing Fig. 7 and Figs. 5 and 6, it can be seen that the flexibility power has the same time change trend as the total active power of RESs in the network. For example, there is only a wind farm in the transmission network, so based on Fig. 7(a), the total flexibility power of EHs located in the transmission network, EH1, and EH2, is the same as the daily wind farm power curve in Fig. 6, except for they are numerically different. This is because based on Eqs. (11) and (19), it is necessary to use sources of flexibility, that is, EHs, to compensate for RESs power fluctuations in different scenarios compared to the scenario with predicted uncertainty values when trying to improve the flexibility of networks. According to Section 4.1.A, the

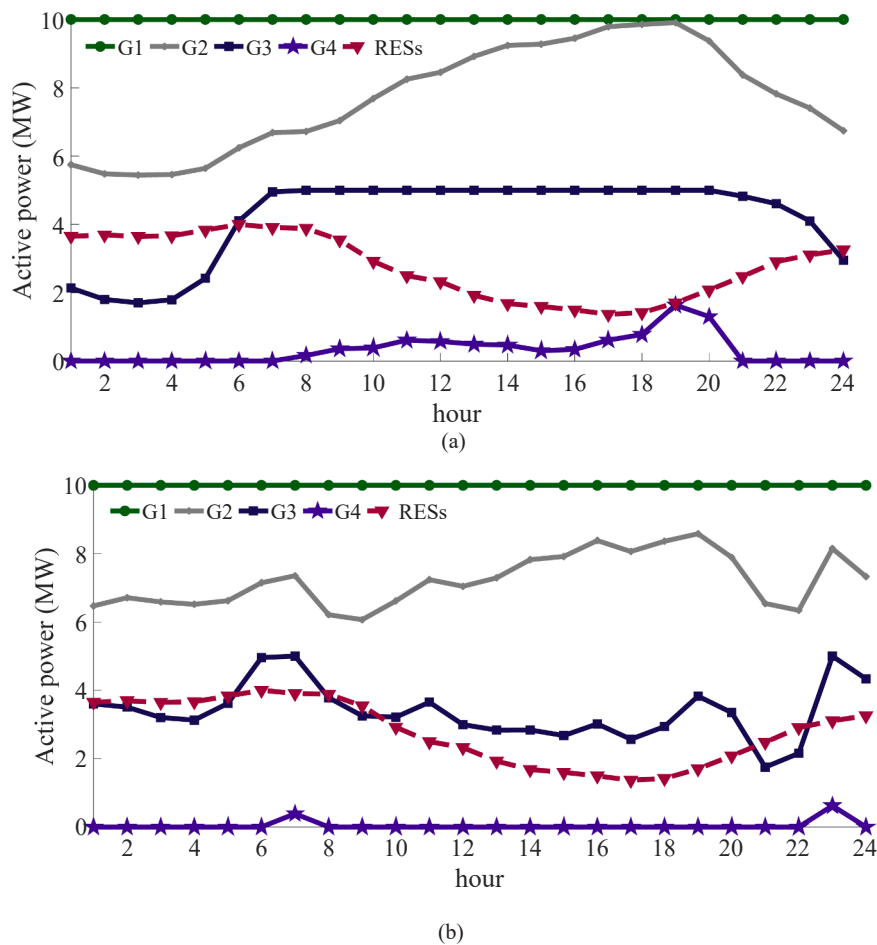


Fig. 6. Expected daily active power curve of generation units and renewable sources in transmission network for case, a) without EHs, b) with EHs.

Table 3

Value of operation indices of different networks.

Case	Without RESs and EHs	Without EHs	With RESs and EHs
Energy loss in TN (MWh)	17.6	15.6	13.7
Energy loss in DN1 (MWh)	2.97	2.73	2.13
Energy loss in DN2 (MWh)	2.97	2.78	2.19
Total energy loss (MWh)	23.54	21.11	18.02
Maximum voltage deviation (p.u.) for TN	0.012	0.011	0.009
Maximum voltage deviation (p.u.) for DN1	0.087	0.079	0.058
Maximum voltage deviation (p.u.) for DN2	0.087	0.082	0.060

standard deviation of uncertainty is considered constant in all hours; hence, with the increase of RESs power generation, it is expected that the amount of RESs power dispersion in different scenarios will increase. So, the flexibility power of EHs increases/decreases with the increase/decrease of RESs power generation. In addition to this, it should be noted that the flexibility resources such as storage devices and responsive loads in Fig. 4 are not turned off for all operating hours. However, in some hours, EVs have zero power, because the number of EVs based on Fig. 3(c) is zero in some hours. Since the total power of RESs in networks is not zero during all operating hours, it is expected that flexible resources such as responsive loads and storage devices in EHs will manage their active power and not shut down.

D) *Evaluation of the economic situation of networks and EHs:* Fig. 8 presents the average daily curve of the energy price of different buses of

transmission and distribution networks, and the daily curve of the flexibility incentive price (FIP) in different networks for two upward and downward modes. Based on Fig. 8(a), in the case without EHs, the energy price for different networks has two levels. In the first level, the price of energy has the lowest value and appears for the hours 1:00–7:00 and 21:00–00:00. In other operating hours, the energy price of the second level appears, which has the highest value. The increase in the price of energy in the hours 8:00–20:00 compared to other operating hours is due to the presence of G4 in the transmission network based on Fig. 6(a). This is because G4 has the highest fuel price, which leads to an increase in energy prices. The energy price for the distribution network is around the same energy price of the transmission network because, according to Fig. 2, distribution networks are connected to transmission network buses. The energy price of the transmission network buses is determined based on the fuel price of the power plants, and there is no source with fuel price in the distribution network. Therefore, the energy price of distribution networks will depend on the fuel price of power plants. In addition, the presence of EHs in the distribution and transmission networks leads to a decrease in the price of energy during all operating hours. Also, the number of hours with high energy prices (second level) decreases because, according to Fig. 6(b), the presence of EHs decreases the number of hours of G4's presence in the transmission network. Fig. 8(b) illustrates the daily curve of FIP. The FIP for upward mode in all networks has two levels. At one level, FIP is high and is low at another level. The high level of FIP by comparing Fig. 8(b) and Fig. 7 corresponds to the high degree of flexibility power of EHs in upward mode. Also, a low FIP case corresponds to the low flexibility power of EHs. It should be noted that according to (Shahidehpour et al., 2003), as the demand for a product increases, the price of the product increases.

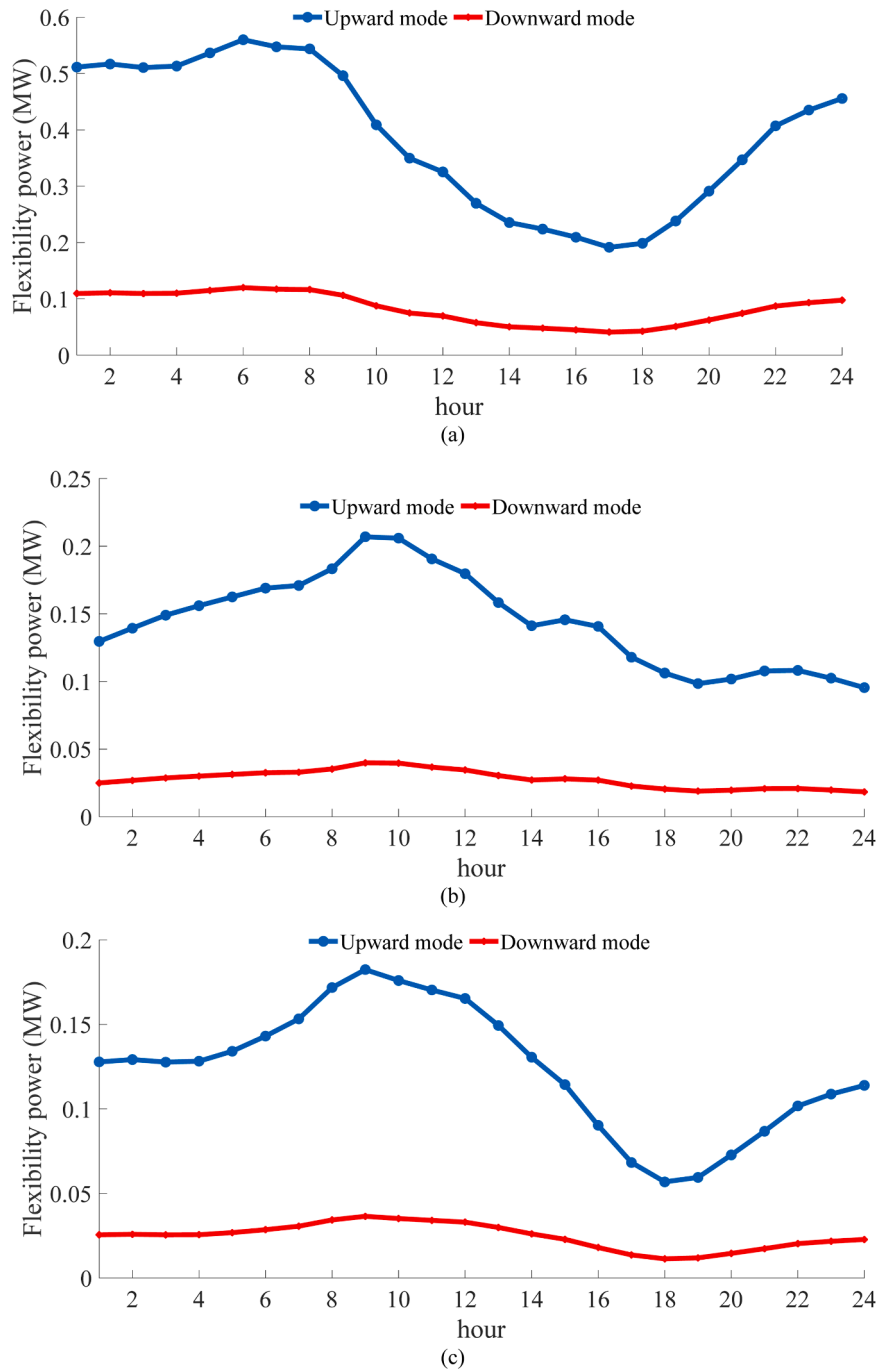


Fig. 7. Expected daily flexibility power curve of EHs in, a) transmission network, b) distribution network 1, c) distribution network 2.

Since the demand for flexibility at any moment for the proposed scheme is equal to the flexibility power of EHs, the high flexibility power of EHs in the upward mode increases the FIP in this mode. Based on Fig. 7, the temporal changes of EHs' flexibility power in the downward mode are insignificant. Therefore, the FIP in downward mode is constant for all networks during all operating hours.

Table 4 lists the economic indicators of networks and EHs for the study cases of a network without EHs and RESs, a network without EHs, and a network with EHs and RESs. The energy cost of TSO (the first row of Eq. (1)) for power flow studies (network without EHs and RESs) is equal to \$17545, and the energy cost of DSO (the second row of Eq. (1)) for DN1 and DN2 is \$2089 for the mentioned study case. With the presence of RESs in the aforementioned networks, the operating cost of

TN is reduced by about 20 % ((17,545–14,034)/17,545)) compared to power flow studies. The operating costs of DN1 and DN2 are also reduced by 23.4 % and 12.15 %, respectively, compared to the case without EHs and RESs. With the presence of EHs in distribution and transmission networks, the energy cost of TN, DN1, and DN2 is reduced by about 26.3 %, 44 %, and 32.6 %, respectively, compared to power flow studies. In this situation, the energy cost of EHs (the third row of Eq. (1)) is equal to –\$1045. Its negative sign means that EHs do not have costs in this situation and they can earn money because of injecting energy into the networks. In addition, in the mentioned conditions, EHs can get \$172 income from flexibility regulation (last row of Eq. (1)) of distribution and transmission networks.

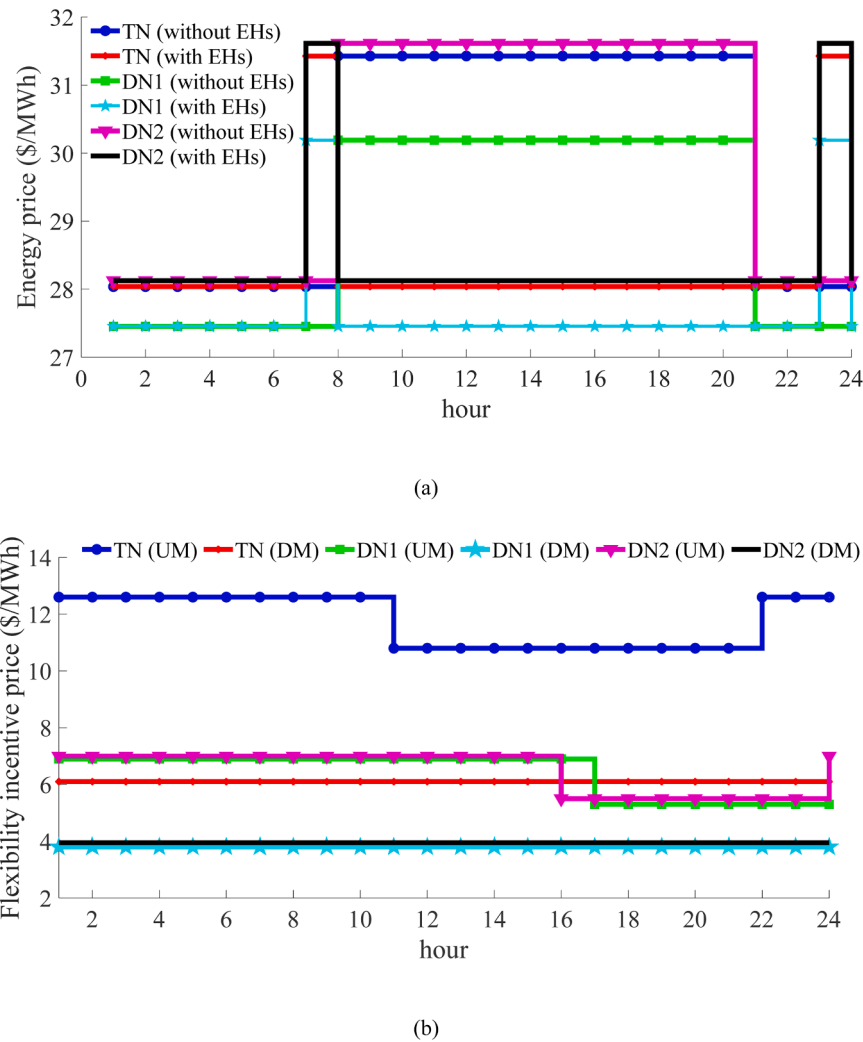


Fig. 8. Expected daily curve of, a) mean of energy price, b) mean of flexibility incentive price for different networks.

Table 4
Value of economic indices in the different networks.

Case	Without RESs and EHS	Without EHS	With RESs and EHS
TSO energy cost (\$)	17545	14034	12931
DSO energy cost (\$) in DN1	2089	1600	1171
DSO energy cost (\$) in DN2	2089	1828	1408
EHS energy cost (\$)	0	0	-1045
EHS flexibility benefit (\$)	0	0	172

4.2. IEEE 30-bus transmission network

In this section, the proposed scheme is applied to the IEEE 30-bus transmission network (Hamidpour et al., 2019). The data of transmission lines, power plants, and peak load for this transmission network is presented in (Hamidpour et al., 2019). The network consists of five IEEE 33-bus distribution networks, whose locations are shown in Fig. 9. The data of these distribution networks, such as the data of EHS and RESs, correspond to DN1 and DN2 in Fig. 2 and Section 4.1.A. In the mentioned transmission network, there are two wind farms with a capacity of 4 MW, whose location is given in Fig. 9. Also, this network has four EHS (Fig. 9), and its data for EH1 and EH2 correspond to Table 2

and Section 4.1.A. Other specifications of this section, such as the permissible ranges of voltage, flexibility tolerance, and standard deviation of uncertainties, are the same as in Section 4.1.A.

The results of this section are provided in Table 5, which shows the economic and operation indicators for cases without RESs and EHS, without EHS, and with RESs and EHS. Based on this table, the highest energy losses, voltage deviations, and operating costs of the networks belong to power flow studies (the case without RESs and EHS). This is because, in this situation, the consumers are supplied only by the power plant. However, its lowest value has appeared in the case of the presence of RESs and EHS so that the total energy loss, voltage deviation in the transmission network and all distribution networks, and the operating cost of the transmission network and distribution networks in this case compared to the case without EHS and RESs for 100 % flexibility (flexibility tolerance is zero) are reduced by 26.2 %, 30.8 %, 38 %, 19.5 %, and 48.1 %, respectively. In this situation, considering 100 % flexibility for distribution and transmission networks, the energy income of EHS is \$2681 (energy cost of EHS is -\$2681). Also, in this case, EHS can earn \$569.3 from regulating the flexibility of networks. In the following, it should be noted that with the increase of flexibility tolerance (decrease in the importance of flexibility based on Eqs. (11) and (19) in the proposed scheme), the situation of economic and operation indicators in distribution and transmission networks are more improved than in power flow studies. For example, if the importance of flexibility is reduced, storage devices and responsive loads are turned off at certain hours to reduce the operating cost of networks. That is, they are in

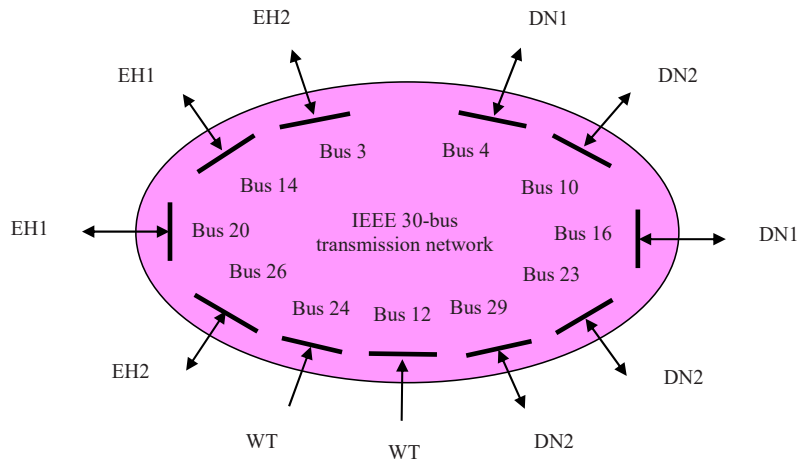


Fig. 9. IEEE 30-bus transmission network (Hamidpour et al., 2019) including sources, EHs, and distribution network.

Table 5
Value of economic indices and calculation time in the different networks.

Case	Without RESs and EHs	Without EHs	With RESs and EHs for β_F (p.u.)				
			0	0.05	0.10	0.15	0.20
Energy loss in TN (MWh)	151.2	134.7	111.4	110.6	109.2	107.6	105.9
Energy loss in DNs (MWh)	14.2	12.8	10.6	10.3	9.8	9.2	8.6
Total energy loss (MWh)	165.4	147.5	122	120.9	119	116.8	114.5
Maximum voltage deviation (p.u.) for TN	0.026	0.023	0.018	0.018	0.018	0.017	0.017
Maximum voltage deviation (p.u.) for all DNs	0.087	0.078	0.054	0.053	0.053	0.052	0.052
TSO energy cost (\$)	142760	131542	114985	113751	110795	107367	104082
DSO energy cost (\$) for all DNs	10394	8658	5391	5278	5012	4698	4211
EHs energy cost (\$)	0	0	-2681	-2691	-2734	-2789	-2846
EHs flexibility benefit (\$)	0	0	569.3	519.2	428.3	317.4	186.5
Calculation time (minute)	6.7	7.3	17.1	16.7	16.2	15.2	14.0

charging (discharging) mode during the hours when the price of energy is the cheapest (most expensive). During the hours when the energy price is between the cheapest and the most expensive, the storage devices and DRP can be turned off. This leads to reduced operating costs. Therefore, by reducing the importance of flexibility, the performance of EH elements will be proportional to the improvement of the economic and operation status of networks. In the case of EHs, increasing flexibility tolerance leads to an increase in EHs' income from energy management, but their income from flexibility regulation drops. Yet, according to Table 5, the total income of EHs resulting from energy management and flexibility regulation decreases with the increase of flexibility tolerance. In addition, the calculation time is low for load flow analysis of networks. This time increases about to 9 % $((7.3 - 6.7) \div 6.7)$ if the RESs added to networks. But, it increases by about 109 %–155 % if RESs and EHs based on the model in Section 2 are added to networks. In this case, the calculation time is reduced by increasing flexibility tolerance.

5. Conclusion

This paper presents the simultaneous energy management of distribution and transmission networks in the presence of EHs corresponding to TSO-DSO coordination. In this scheme, the objective function is responsible for minimizing the total energy costs of the aforementioned networks and EHs minus the revenue of the EHs from the flexibility regulation of the networks. This scheme is constrained to optimal power flow equations, the flexibility limit of networks, and the EHs flexibility-operation model. The mentioned scheme has uncertainties relating to the load, renewable power, and mobile storage parameters. The unscented transformation method was adopted to model these

uncertainties. The price of energy and FIP, respectively, are equal to the dual variable of the active power balance and the limitation of the flexibility of the networks. For the simultaneous calculation of these variables and the main variables of the problem, an integrated model for the proposed scheme was obtained from the penalty function method. Eventually, based on the numerical results, it was observed that the sole presence of RESs in the abovementioned networks can enhance the operation and economic status of distribution and transmission networks by 5 %–11 % and 12 %–24 %, respectively, compared to power flow studies. However, the simultaneous presence of EHs and RESs in distribution and transmission networks leads to the improvement of the economic and operation status of networks by 26 %–44 % and 22 %–33 %, respectively. Of course, EHs can extract 100 % flexibility conditions for networks. Also, they can earn significant revenue from flexibility regulation and energy management for power sources, responsive loads, and storage devices.

A comparison was made between different scenarios to analyze the flexibility of the system in the proposed scheme. Although this modeling provides a correct formulation of flexibility, in the real issue of flexibility, it is necessary to compare the results of day-ahead and real-time operations. Therefore, the proposed scheme is considered as a future work considering the mentioned operations. In addition, based on the numerical results, it has been observed that the hubs are able to operate as energy producers in most hours. So, by their presence in the network, they can feed a part of the consumed energy of the network. Now, if the case of an N-1 event, hubs can improve the reliability of the network by supplying a percentage of the demand energy. However, this issue was included in the proposed scheme as a future work. In this paper, electric energy storage in the form of a battery was utilized. Despite high power density and efficiency of a battery, its installation cost is high and its

lifespan is low. Installing its high capacity has different complications. Nonetheless, hydrogen storage and compressed air can be created in high capacity, their useful lifespan is high, and their installation cost is lower than batteries. They also have acceptable work efficiency. Therefore, the proposed scheme with hydrogen storage and compressed air was included as a future work. The suggested scheme is based on the optimal operation of hubs. Yet, one of the best solutions to improve the economic and technical condition of the network is using a certain number of hubs, including the planning of the hubs, the location of the hubs, and the measurement of resources and storage devices in the hubs. This issue is considered for future work.

CRedit authorship contribution statement

Leila Bagherzadeh: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Resources, Software, Visualization, Writing – original draft, Writing – review & editing. **Innocent Kamwa:** Investigation, Project administration, Supervision, Writing – review & editing. **Yahya Z.Alharthi:** Investigation, Validation.

Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability

The data that has been used is confidential.

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